Student Name: _____

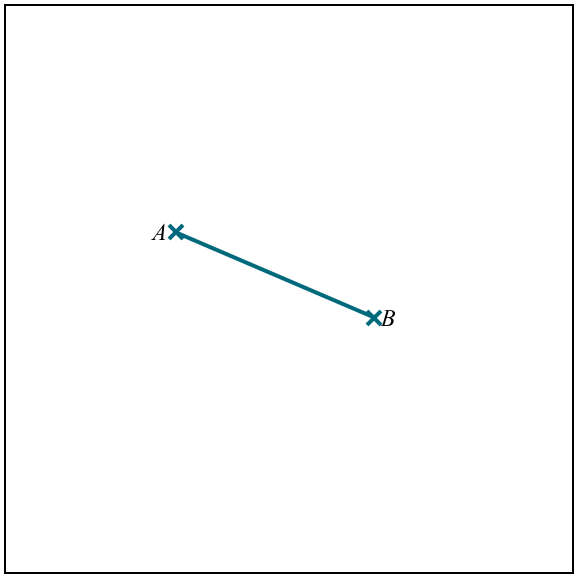
Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **25**

Instructor Name: **Bodanske, Daniel**

Question 1 of 25

Use a compass and ruler to construct the perpendicular bisector of the line segment $\overline{AB}$.

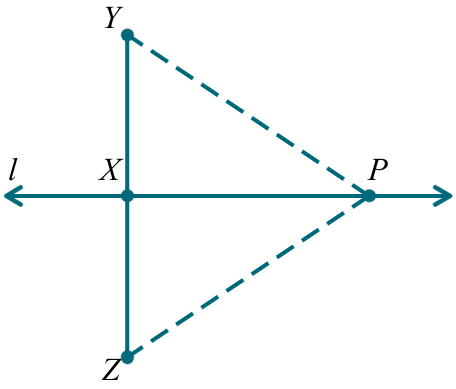


Question 2 of 25

Use the given information to prove the following theorem.

If a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment.

We let P be any point on line l, but different from point X.



Given: $\overline{PX}$ is the perpendicular bisector of $\overline{YZ}$

Prove: $YP = ZP$

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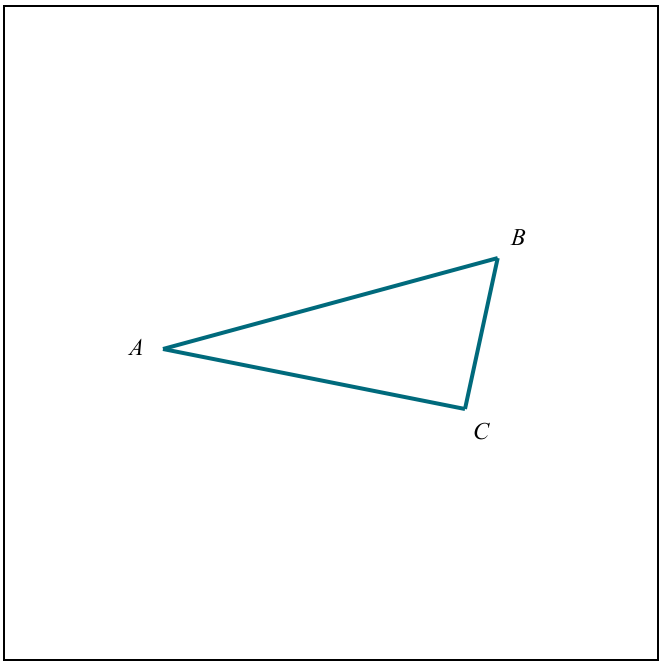
Question 3 of 25

Use a compass and ruler to draw a circle with center A and a radius of 6 units.

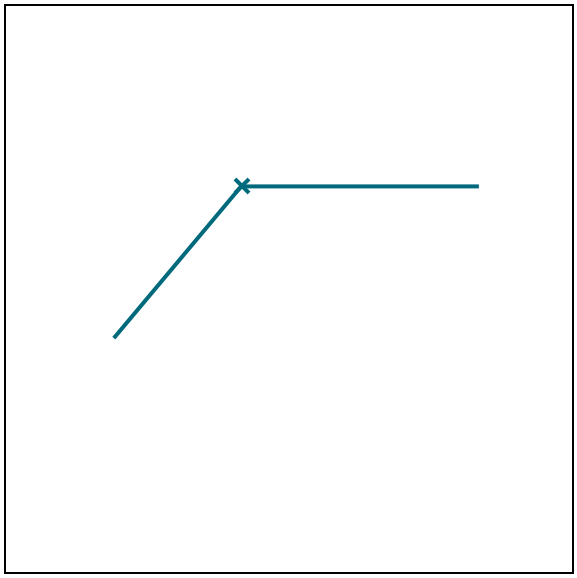
Question 4 of 25

Use a compass and ruler to circumscribe a circle about the triangle.



Question 5 of 25

Use a compass and ruler to construct the bisector of the angle given below.



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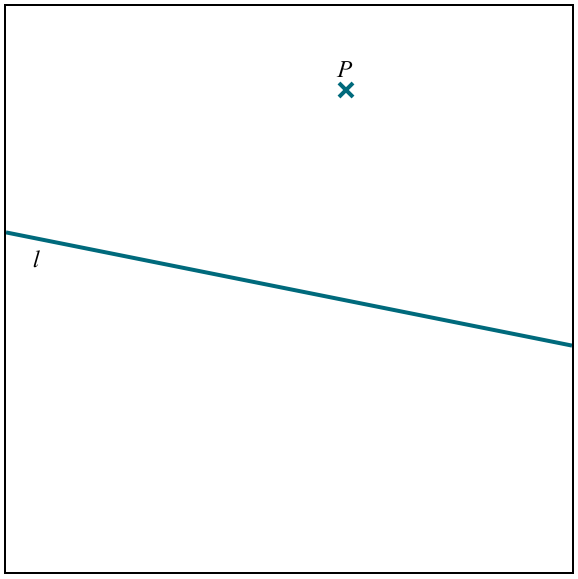
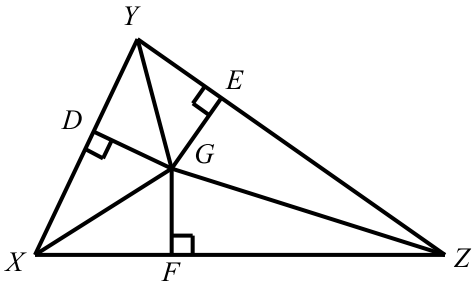
Question 6 of 25

Use a compass and ruler to construct the line perpendicular to the line l through the point P.



Question 7 of 25

The angle bisectors of $\triangle XYZ$ are $\overline{XG}$, $\overline{YG}$, and $\overline{ZG}$. They meet at a single point G.
(In other words, G is the incenter of $\triangle XYZ$.)
Suppose $FG = 18$, $YG = 27$, $m\angle DXF = 36°$, and $m\angle EYG = 58°$.
Find the following measures.
Note that the figure is not drawn to scale.



$m\angle EYD =$ _____ °

$DG =$ _____

$m\angle EZG =$ _____ °

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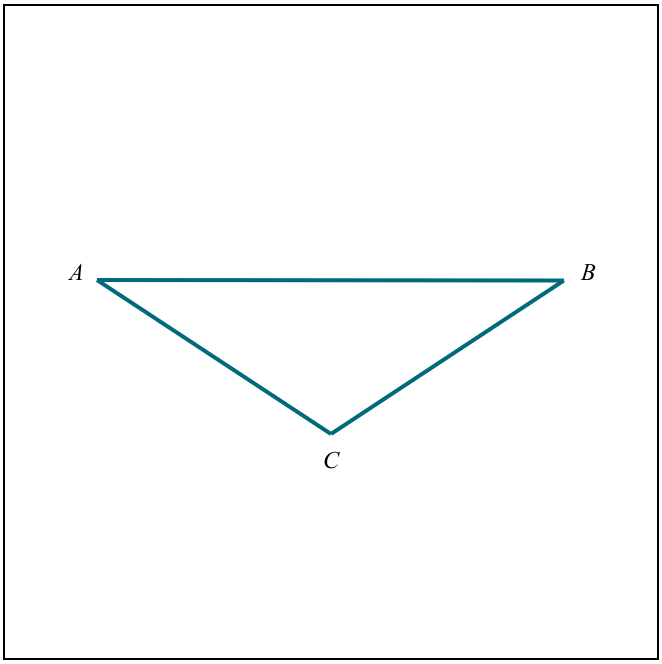
Question 8 of 25

Use a compass and ruler to inscribe a circle in the triangle.



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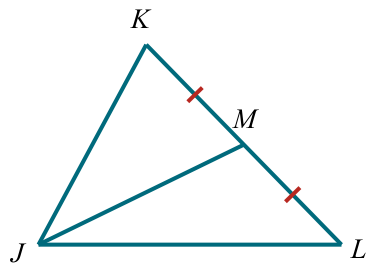
Question 9 of 25

Classify the segment in each part below.
Only use the information given in the figure.
Do not assume measures are equal unless the markings indicate they are.

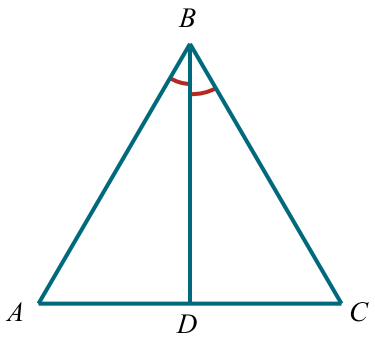
(a) Select *all* that describe $\overline{JM}$.

- [ ] Perpendicular bisector of $\overline{KL}$
- [ ] Angle bisector of $\angle J$
- [ ] Median of $\triangle JKL$
- [ ] Altitude of $\triangle JKL$
- [ ] None of the above



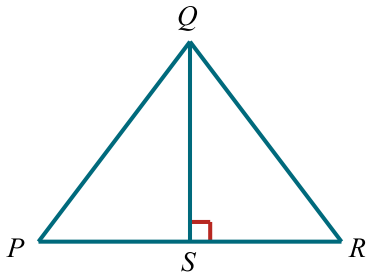
(b) Select *all* that describe $\overline{BD}$.

- [ ] Perpendicular bisector of $\overline{AC}$
- [ ] Angle bisector of $\angle B$
- [ ] Median of $\triangle ABC$
- [ ] Altitude of $\triangle ABC$
- [ ] None of the above



(c) Select *all* that describe $\overline{QS}$.

- [ ] Perpendicular bisector of $\overline{PR}$
- [ ] Angle bisector of $\angle Q$
- [ ] Median of $\triangle PQR$
- [ ] Altitude of $\triangle PQR$
- [ ] None of the above



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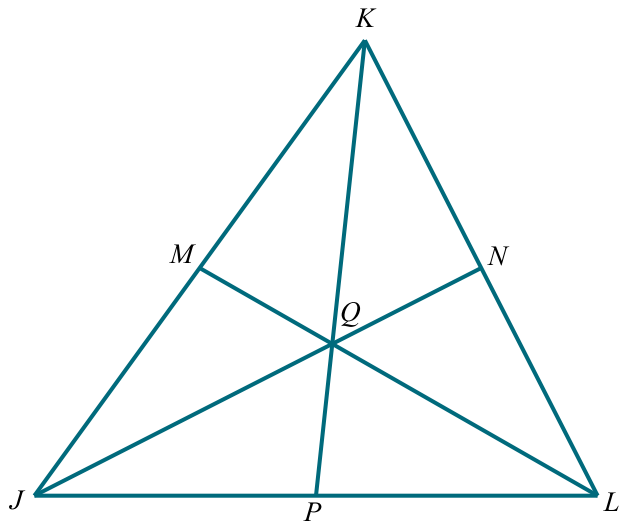
The medians of $\triangle JKL$ are $\overline{JN}$, $\overline{KP}$, and $\overline{LM}$. They meet at a single point Q.
(In other words, Q is the centroid of $\triangle JKL$.)

Suppose $QP = 11$, $QM = 10$, and $JN = 39$.
Find the following lengths.
Note that the figure is not drawn to scale.



$LM =$ _____

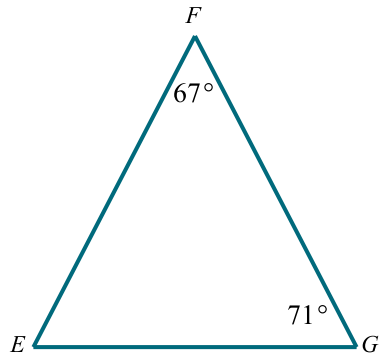
$JQ =$ _____

$KQ =$ _____

Question 11 of 25

Order the side lengths EF, FG, and GE from least to greatest.

(Note that the figure is not drawn to scale.)



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Question 12 of 25

For each sentence, determine whether it's a statement.

Sentence	Statement?	
	Yes	No
Yowza!	○	○
Do your homework.	○	○
Who is your best friend?	○	○
$40 \div 8 = 7$	○	○
Vietnam has the largest population in the world.	○	○

Question 13 of 25

Consider statements p and q.

p: We sell carrots at the market.
q: We sell berries at the market.

For parts (a) and (b), fill in the symbolic form. For part (c), choose the descriptive form.

Descriptive form	Symbolic form
(a) We sell carrots at the market or we sell berries at the market.	
(b) We sell carrots at the market and we sell berries at the market.	
(c) (Choose a description) • We sell carrots at the market. • We do not sell carrots at the market. • We do not sell berries at the market.	$\sim q$

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Question 14 of 25

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The table below gives the material of each item in Jane's jewelry box.
These are the only items in Jane's jewelry box.

Item	Material
Bracelet	Gold
Ring	Silver
Pendant	Platinum
Necklace	Gold
Hairpin	Copper

A conditional statement is given below. Give the contrapositive, converse, and inverse of the statement. Also use the table above to decide if each statement is true or false.

Assume that any item mentioned comes from Jane's jewelry box.

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Given statement	If an item is a bracelet, then the item is made from gold.	○ True ○ False
Contrapositive	○ If an item is a bracelet, then the item is a bracelet. ○ If an item is a bracelet, then the item is not a bracelet. ○ If an item is a bracelet, then the item is made from gold. ○ If an item is a bracelet, then the item is not made from gold. ○ If an item is not a bracelet, then the item is a bracelet. ○ If an item is not a bracelet, then the item is not a bracelet. ○ If an item is not a bracelet, then the item is made from gold. ○ If an item is not a bracelet, then the item is not made from gold. ○ If an item is made from gold, then the item is a bracelet. ○ If an item is made from gold, then the item is not a bracelet. ○ If an item is made from gold, then the item is made from gold. ○ If an item is made from gold, then the item is not made from gold. ○ If an item is not made from gold, then the item is a bracelet. ○ If an item is not made from gold, then the item is not a bracelet. ○ If an item is not made from gold, then the item is made from gold. ○ If an item is not made from gold, then the item is not made from gold.	○ True ○ False
Converse	○ If an item is a bracelet, then the item is a bracelet. ○ If an item is a bracelet, then the item is not a bracelet. ○ If an item is a bracelet, then the item is made from gold. ○ If an item is a bracelet, then the item is not made from gold. ○ If an item is not a bracelet, then the item is a bracelet. ○ If an item is not a bracelet, then the item is not a bracelet. ○ If an item is not a bracelet, then the item is made from gold. ○ If an item is not a bracelet, then the item is not made from gold. ○ If an item is made from gold, then the item is a bracelet. ○ If an item is made from gold, then the item is not a bracelet. ○ If an item is made from gold, then the item is made from gold. ○ If an item is made from gold, then the item is not made from gold. ○ If an item is not made from gold, then the item is a bracelet. ○ If an item is not made from gold, then the item is not a bracelet. ○ If an item is not made from gold, then the item is made from gold. ○ If an item is not made from gold, then the item is not made from gold.	○ True ○ False
Inverse	○ If an item is a bracelet, then the item is a bracelet.	○ ○

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	○ If an item is a bracelet, then the item is not a bracelet. ○ If an item is a bracelet, then the item is made from gold. ○ If an item is a bracelet, then the item is not made from gold. ○ If an item is not a bracelet, then the item is a bracelet. ○ If an item is not a bracelet, then the item is not a bracelet. ○ If an item is not a bracelet, then the item is made from gold. ○ If an item is not a bracelet, then the item is not made from gold. ○ If an item is made from gold, then the item is a bracelet. ○ If an item is made from gold, then the item is not a bracelet. ○ If an item is made from gold, then the item is made from gold. ○ If an item is made from gold, then the item is not made from gold. ○ If an item is not made from gold, then the item is a bracelet. ○ If an item is not made from gold, then the item is not a bracelet. ○ If an item is not made from gold, then the item is made from gold. ○ If an item is not made from gold, then the item is not made from gold.	True False

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Question 15 of 25

Show that each statement is false by providing a counterexample.

(a) If the area of a rectangle is 100, then the length is 25 and the width is 4.

Counterexample: length $=$ width

(b) If $\angle 1$ and $\angle 2$ are supplementary angles, then one of them must be acute.

Counterexample: $m\angle 1 = m\angle 2$

(c) If the length of $\overline{AC}$ is 20 and point B lies on $\overline{AC}$, then $AB = 10$ and $BC = 10$.

Counterexample: $AB = BC$

(d) If the measures of $\angle R$, $\angle S$, and $\angle T$ sum to $180°$, then one of the angles must be obtuse.

Counterexample: $m\angle R = m\angle S$

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Question 16 of 25

Complete the following indirect proof (proof by contradiction).

Given: $\triangle ABC$

Prove: $\triangle ABC$ has at least one angle with measure $60°$ or less

Proof:

First, we assume that this conclusion is false. In other words, we assume that the contrary statement " $\triangle ABC$ has _____ {(a) no angle, (b) at least one angle, (c) exactly one angle, (d) exactly two angles, (e) all three angles} with measure _____ {(a) $60°$ or greater, (b) $60°$ or less, (c) less than $60°$, (d) greater than $60°$}" is _____ {(a) true, (b) false}.

This is equivalent to the following three statements:

(1)$m\angle A$ ___ {(a) $\geq$, (b) $\leq$, (c) $=$, (d) $>$, (e) $<$} ____°

(2)$m\angle B$ ___ {(a) $\geq$, (b) $\leq$, (c) $=$, (d) $>$, (e) $<$} ____°

(3)$m\angle C$ ___ {(a) $\geq$, (b) $\leq$, (c) $=$, (d) $>$, (e) $<$} ____°

Using (1)-(3) and addition properties of inequalities, we conclude that $m\angle A + m\angle B + m\angle C$ ___ {(a) $\geq$, (b) $\leq$, (c) $=$, (d) $>$, (e) $<$} $180°$

But this contradicts the _____ {(a) Definition of Right Angles, (b) Angle Addition Property, (c) Triangle Angle-Sum Property, (d) Exterior Angles Property, (e) Linear Pair Property}, which states that $m\angle A + m\angle B + m\angle C$ ___ {(a) $\geq$, (b) $\leq$, (c) $=$, (d) $>$, (e) $<$} $180°$

Therefore, the assumption made is _____ {(a) true, (b) false}, and the statement "$\triangle ABC$ has at least one angle with measure $60°$ or greater" is _____ {(a) true, (b) false}.

Question 17 of 25

In each figure below, the segments on the sides can be moved.
Create the following triangles, if possible.

First figure: A triangle with side lengths of 5, 7, and 8.

Second figure: A triangle with side lengths of 2, 4, and 7.

If there is no triangle possible with the given side lengths, click on "No triangle possible".

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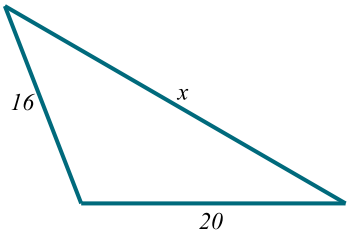
Question 18 of 25

For each set of three lengths, determine if they can be the side lengths of a triangle.

Lengths	Can be side lengths of a triangle	Cannot be side lengths of a triangle
$16, 5, 18$	○	○
$8, 9, 7$	○	○
$8, 25, 17$	○	○
$9.8, 14.0, 23.0$	○	○

Question 19 of 25

The side lengths of a triangle are 16, 20, and x units.



(The figure above gives only one possible configuration for the triangle.)

Determine the possible values of x.
Write your answer as an inequality.
Use x only once in your inequality.

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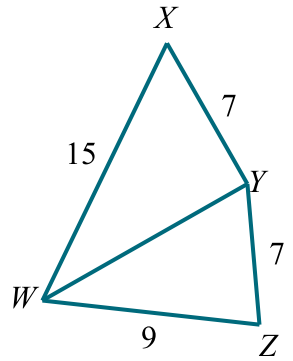
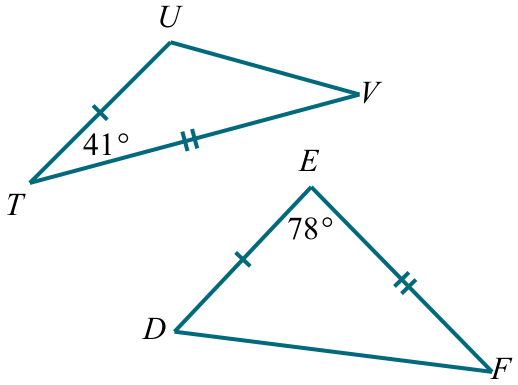
Question 20 of 25

Compare the given measures by choosing $<$, $>$, or $=$.
If there is not enough information, select "Cannot be determined".

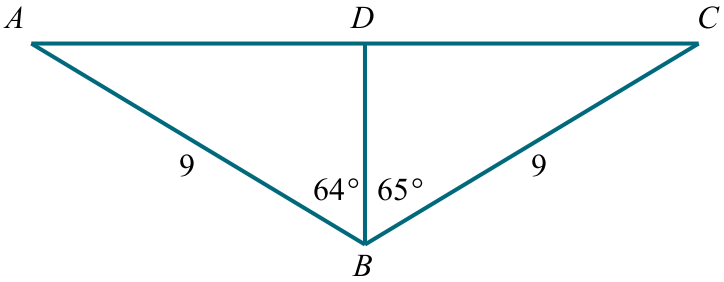
If necessary, you may learn what the markings on a figure indicate.



- ○ $m\angle WYX$ ($<$/$>$/$=$) $m\angle WYZ$
- ○ Cannot be determined



- ○ UV ($<$/$>$/$=$) DF
- ○ Cannot be determined



- ○ AD ($<$/$>$/$=$) CD
- ○ Cannot be determined

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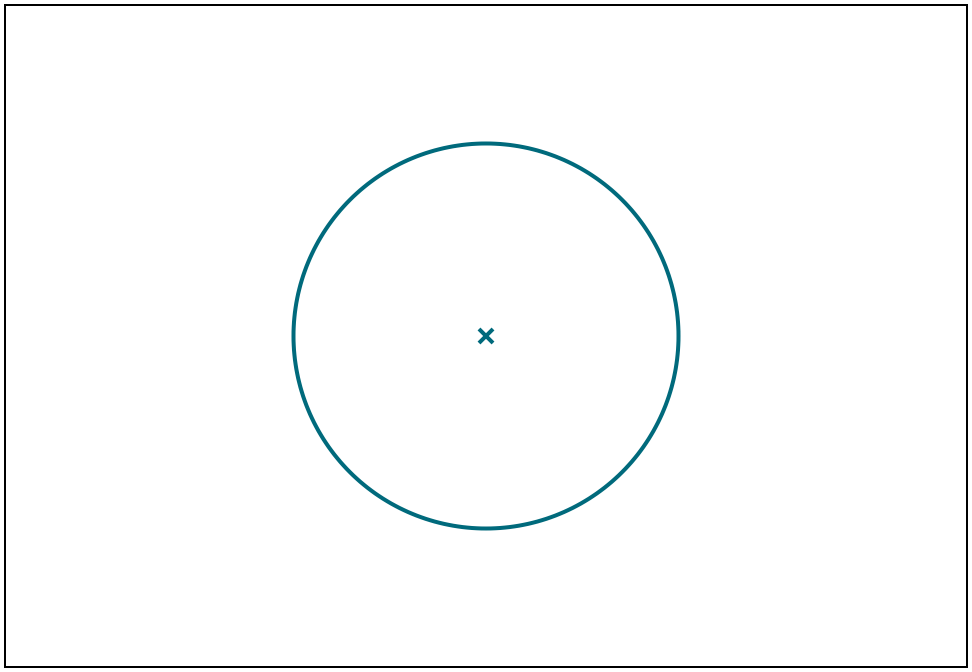
Question 21 of 25

A circle and its center are shown in the figure below.

Use a compass and ruler to inscribe a regular hexagon in the circle.



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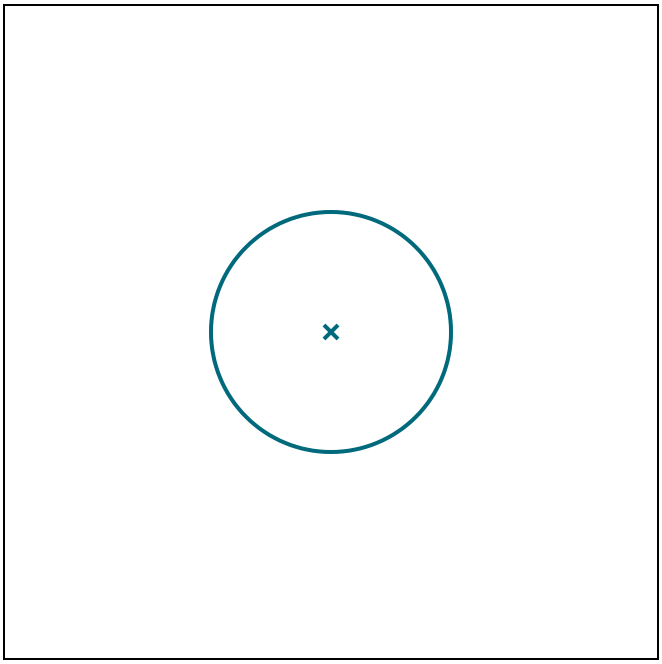
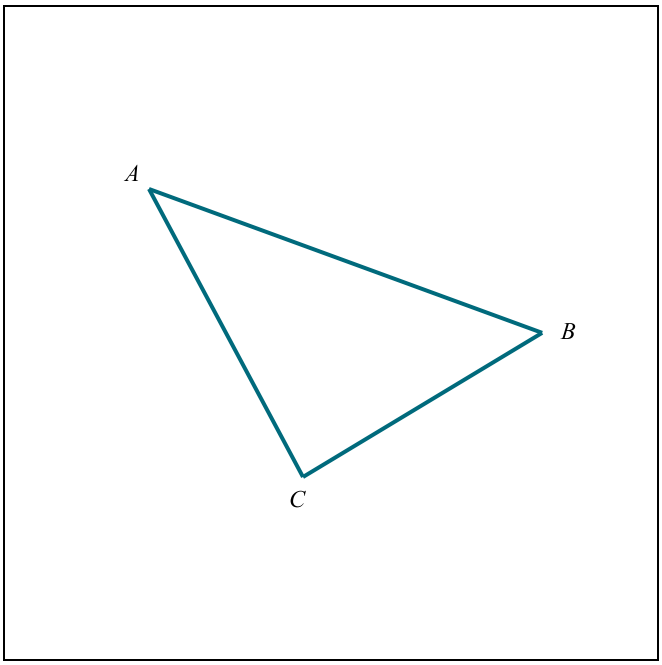
A circle and its center are shown in the figure below.

Use a compass and ruler to inscribe a square in the circle.



Question 23 of 25

Use a compass and ruler to inscribe a circle in the triangle.



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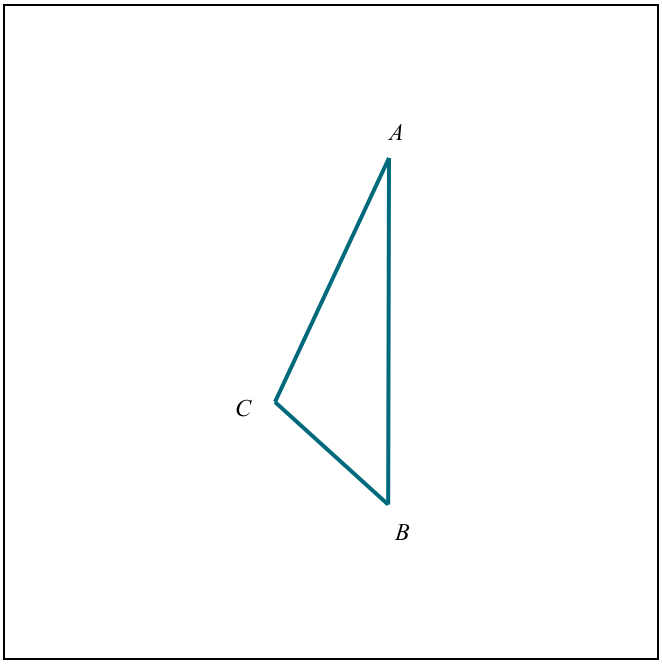
Use a compass and ruler to circumscribe a circle about the triangle.



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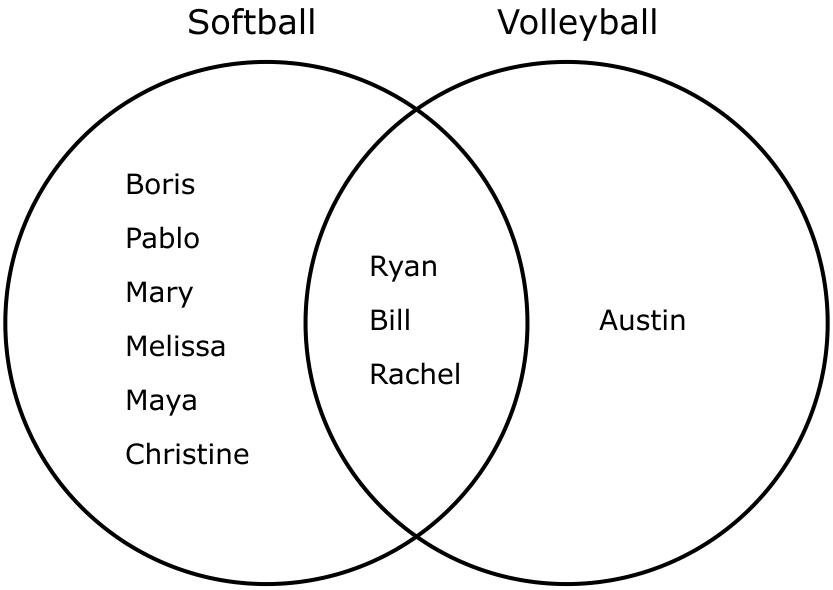
Question 25 of 25

The Venn diagram shows the memberships for the Softball Club and the Volleyball Club.
Use the diagram to answer the questions below.



(a) How many people play Softball?

(b) How many people play Volleyball but not Softball?

(c) How many people play both Softball and Volleyball?

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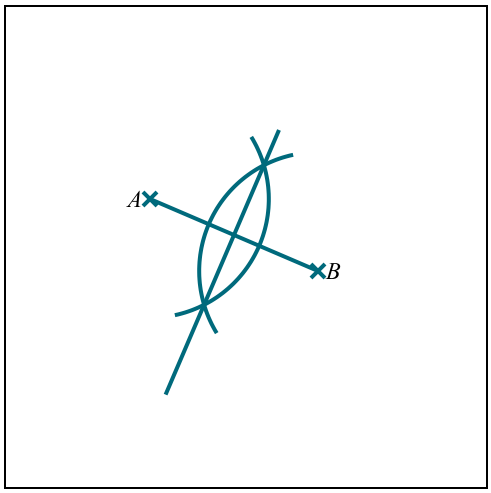
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **25**

Question 1 of 25



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Question 2 of 25

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For this proof, you should be familiar with the following definitions and properties.

- Definition of segment bisector
- Definition of perpendicular bisector
- Definition of perpendicular
- Definition of equidistant
- Definition of right angle
- Reflexive property
- Triangle congruence properties
- CPCTC Property
- Definition of congruent

We are proving the following theorem.

> *If a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment.*

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Understanding the theorem:

Refer to Figure 1. $\overline{PX}$ is the $\perp$ bisector of $\overline{YZ}$.
Point P represents *any* point on line l other than X. The theorem states that P must be equidistant from the endpoints Y and Z.

So, to prove the theorem, we must prove that $YP = ZP$.

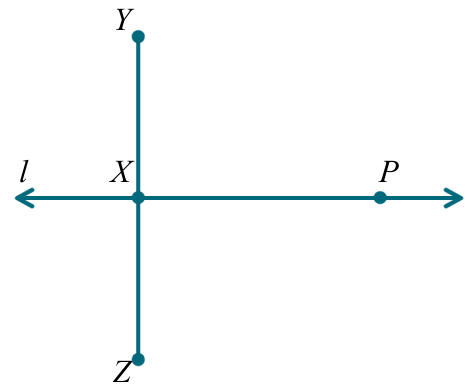

Figure 1

Proving the theorem:

We start by drawing $\overline{PY}$ and $\overline{PZ}$. Notice that this creates $\triangle YXP$ and $\triangle ZXP$.

We can prove $YP = ZP$ as follows.

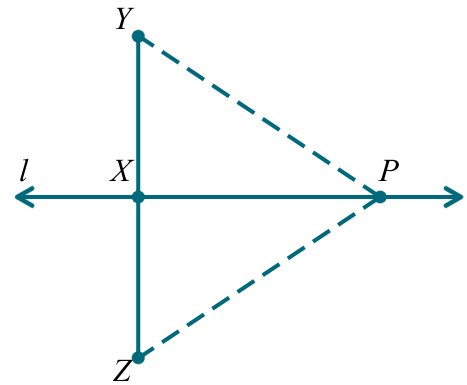

Figure 2

1. We are given that $\overline{PX}$ is the $\perp$ bisector of $\overline{YZ}$.
2. By the definition of a perpendicular bisector, statement 1 implies that $\overline{YX} \cong \overline{ZX}$. This gives us a pair of congruent sides.
3. By the definition of a perpendicular bisector, statement 1 implies that $\angle YXP$ is a right angle.
4. Again, by the definition of a perpendicular bisector, statement 1 implies that $\angle ZXP$ is a right angle.
5. All right angles are congruent, therefore $\angle YXP \cong \angle ZXP$. This gives us a pair of congruent angles.
6. By the reflexive property of congruence, $\overline{PX}$ is congruent to $\overline{PX}$. This gives us a pair of congruent sides since $\overline{PX}$ is a part of both triangles.
7. Now we have $\triangle YXP \cong \triangle ZXP$ by the SAS Congruence Property.
8. So, $\overline{YP} \cong \overline{ZP}$ by the CPCTC Property.
9. Finally, $YP = ZP$ by the Definition of Congruent Segments.

It is important to note the following:
In the proof, we do not specify how far P is from X. (The figure gives only one possible example.) So, we showed that any point P on line l, other than the point X, is equidistant from Y and Z. The point X is the midpoint of segment $\overline{YZ}$, so it is also equidistant from Y and Z. Thus, the theorem is true in general (for all points on line l).

Here is one possible answer.

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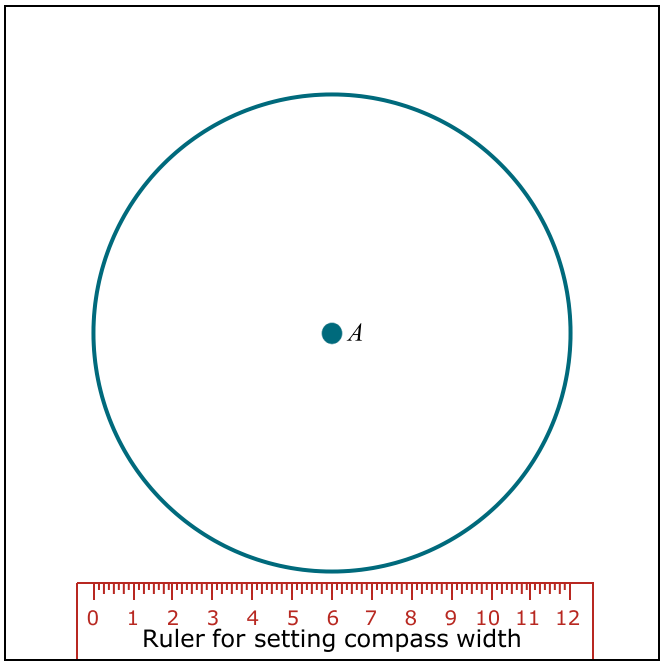
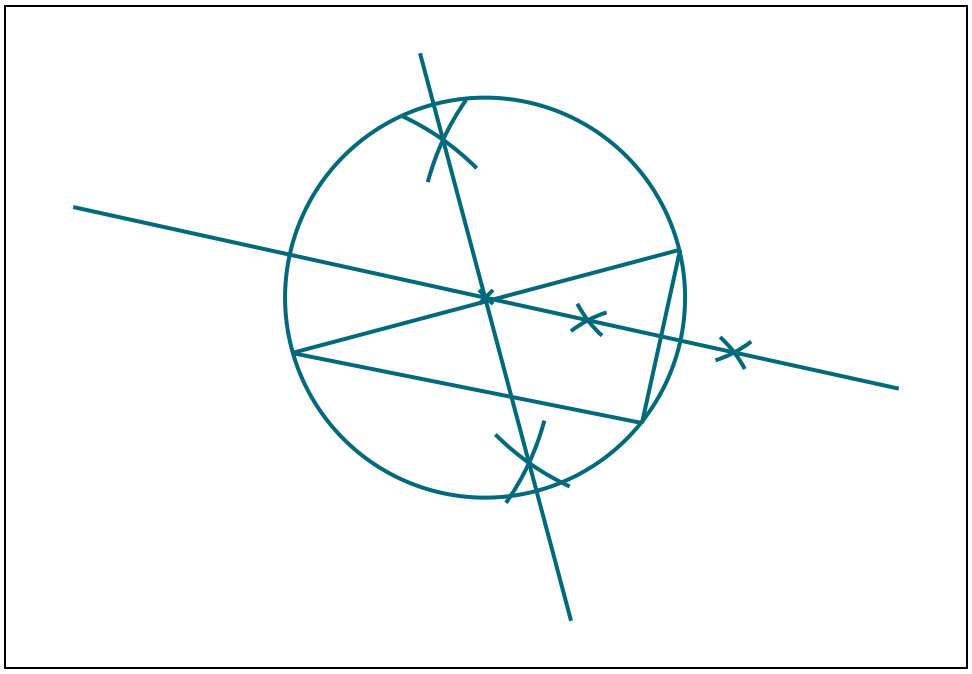
Question 3 of 25



Question 4 of 25



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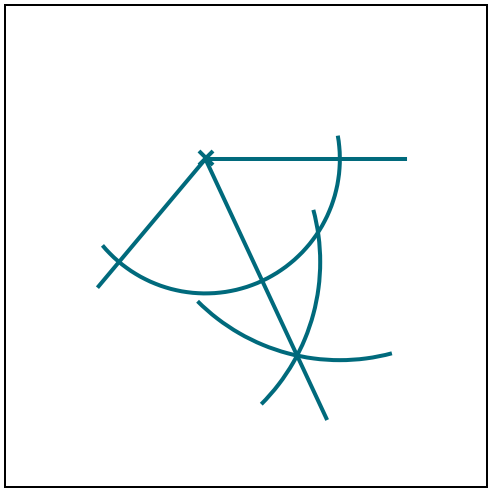
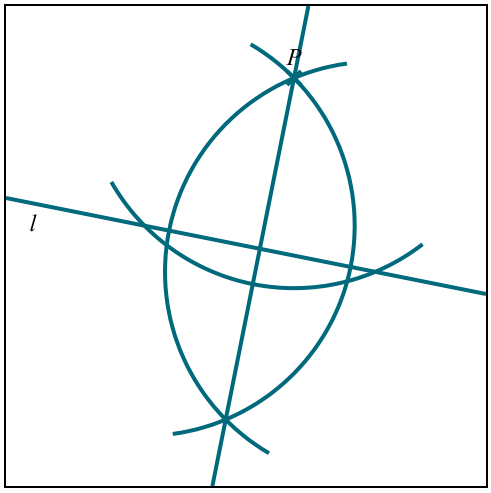
Question 5 of 25



Question 6 of 25



Question 7 of 25

$m\angle EYD = 116°$

$DG = 18$

$m\angle EZG = 14°$

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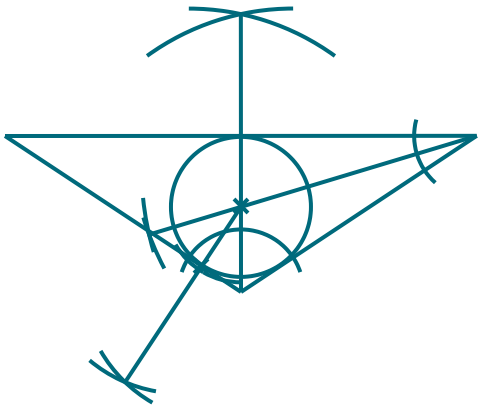
Question 8 of 25



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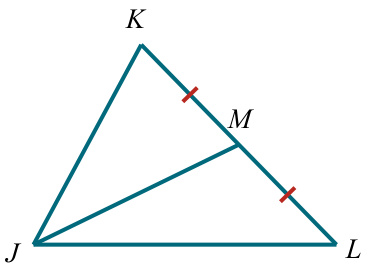
Question 9 of 25

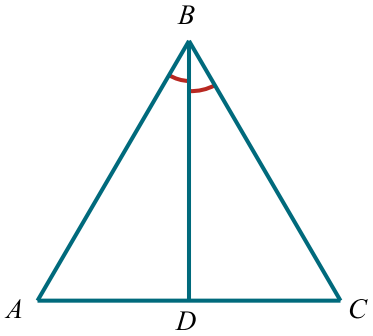
(a) Select *all* that describe $\overline{JM}$.

- [ ] Perpendicular bisector of $\overline{KL}$
- [ ] Angle bisector of $\angle J$
- [x] Median of $\triangle JKL$
- [ ] Altitude of $\triangle JKL$
- [ ] None of the above



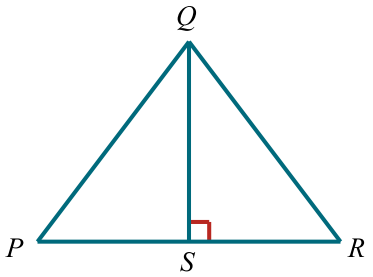
(b) Select *all* that describe $\overline{BD}$.

- [ ] Perpendicular bisector of $\overline{AC}$
- [x] Angle bisector of $\angle B$
- [ ] Median of $\triangle ABC$
- [ ] Altitude of $\triangle ABC$
- [ ] None of the above



(c) Select *all* that describe $\overline{QS}$.

- [ ] Perpendicular bisector of $\overline{PR}$
- [ ] Angle bisector of $\angle Q$
- [ ] Median of $\triangle PQR$
- [x] Altitude of $\triangle PQR$
- [ ] None of the above



Question 10 of 25

$LM = 30$

$JQ = 26$

$KQ = 22$

Question 11 of 25

$FG < GE < EF$

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Question 12 of 25

Sentence	Statement?	
	Yes	No
Yowza!	○	○
Do your homework.	○	○
Who is your best friend?	○	○
$40 \div 8 = 7$	○	○
Vietnam has the largest population in the world.	○	○

Question 13 of 25

Descriptive form	Symbolic form
(a) We sell carrots at the market or we sell berries at the market.	$p \vee q$
(b) We sell carrots at the market and we sell berries at the market.	$p \wedge q$
(c) (Choose a description) • We do not sell berries at the market.	$\sim q$

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Question 14 of 25

Given statement	If an item is a bracelet, then the item is made from gold.	◉ True ○ False
Contrapositive	If an item is not made from gold, then the item is not a bracelet.	◉ True ○ False
Converse	If an item is made from gold, then the item is a bracelet.	○ True ◉ False
Inverse	If an item is not a bracelet, then the item is not made from gold.	○ True ◉ False

Question 15 of 25

(a) Counterexample: length $= 50$, width $= 2$

(b) Counterexample: $m\angle 1 = 90°$, $m\angle 2 = 90°$

(c) Counterexample: $AB = 13$, $BC = 7$

(d) Counterexample: $m\angle R = 30°$, $m\angle S = 70°$, $m\angle T = 80°$

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Question 16 of 25

Proof:
First, we assume that this conclusion is false. In other words, we assume that the contrary statement " $\triangle ABC$ has **all three angles** with measure **greater than 60°"** is **true**.

This assumption is equivalent to the following three statements:
(1) $m\angle A > 60°$, and
(2) $m\angle B > 60°$, and
(3) $m\angle C > 60°$.

Using (1)-(3) and addition properties of inequalities, we conclude that $m\angle A + m\angle B + m\angle C > 180°$

But this contradicts the **Triangle Angle-Sum Property**, which states that $m\angle A + m\angle B + m\angle C = 180°$

Therefore, the assumption made is **false**, and the statement "$\triangle ABC$ has at least one angle with measure $60°$ or greater" is **true**.

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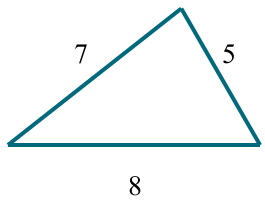

No triangle possible

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Question 18 of 25

Lengths	Can be side lengths of a triangle	Cannot be side lengths of a triangle
16, 5, 18	◉	○
8, 9, 7	◉	○
8, 25, 17	○	◉
9.8, 14.0, 23.0	◉	○

Question 19 of 25

$4 < x < 36$

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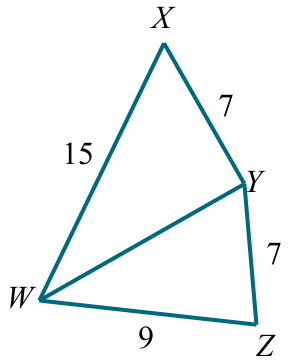
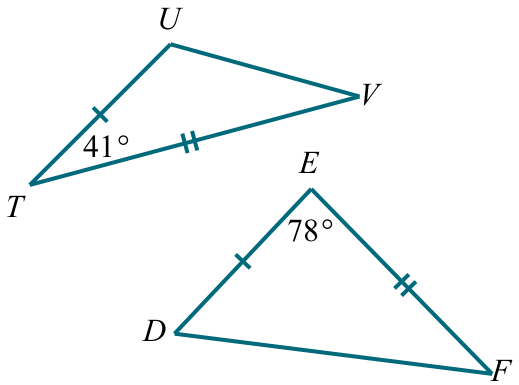
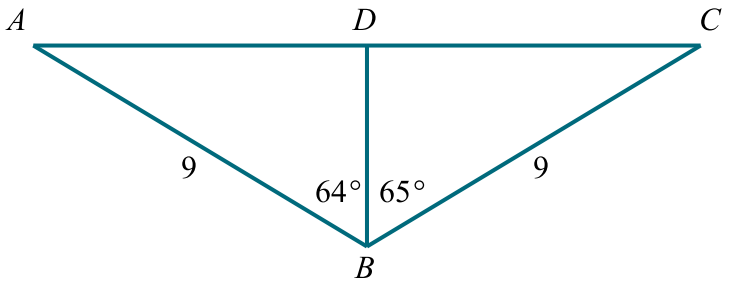
Question 20 of 25



- (●) $m\angle WYX > m\angle WYZ$
- ( ) Cannot be determined



- (●) $UV < DF$
- ( ) Cannot be determined



- (●) $AD < CD$
- ( ) Cannot be determined

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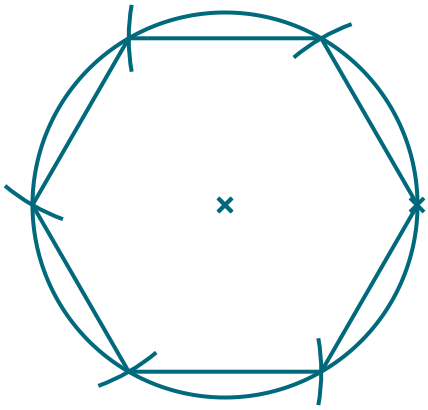
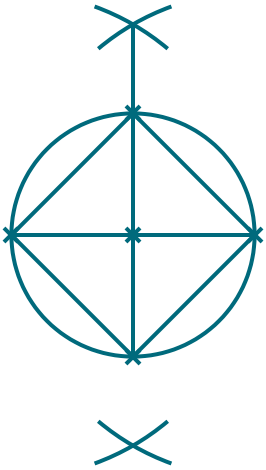
Question 21 of 25



Question 22 of 25



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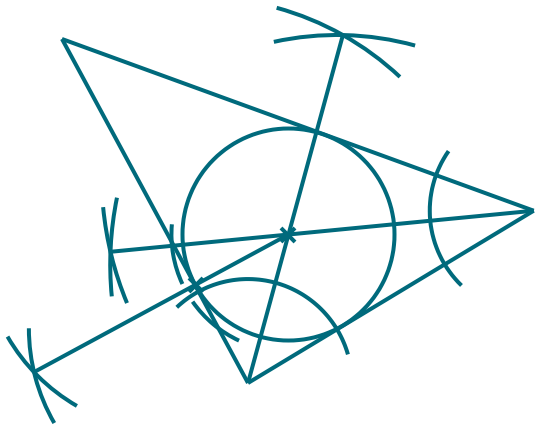
Question 23 of 25



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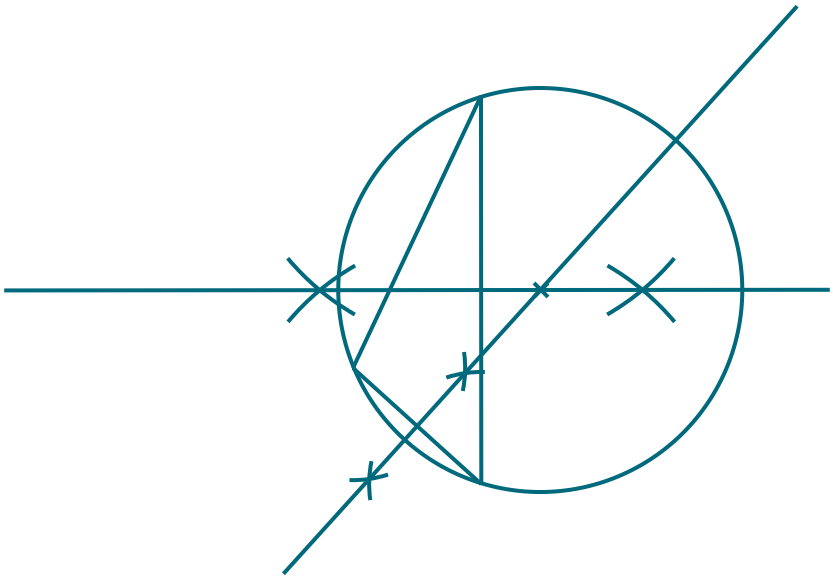

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Question 25 of 25

(a) How many people play Softball?

9

(b) How many people play Volleyball but not Softball?

1

(c) How many people play both Softball and Volleyball?

3

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